Infection, Genetics and Evolution 107 (2023) 105398

Contents lists available at ScienceDirect

Infection, Genetics and Evolution

journal homepage: www.elsevier.com/locate/meegid

Comparative genome analysis of *Streptococcus* strains to identify virulent genes causing neonatal meningitis

Jasmine Arya[a,1], Divya Sharma[a,1], Dev Kumar[a], Ritu Jakhar[a], Alka Khichi[b], Mehak Dangi[a,*], Anil Kumar Chhillar[b,*]

[a] *Centre for Bioinformatics, Maharshi Dayanand University, Rohtak, Haryana, India*
[b] *Centre for Biotechnology, Maharshi Dayanand University, Rohtak, Haryana, India*

A R T I C L E I N F O

Keywords:
Genomic comparison
Non-pathogenic streptococcus strain
Streptococcus agalactiae
Putative virulent factors
Neonatal meningitis
Group B Streptococcus

A B S T R A C T

Aim: To determine *Streptococcus agalactiae* genes responsible for causing neonatal meningitis.
Background: Streptococcus agalactiae strain 2603 V/R is causative agent of neonatal meningitis, maternal infection and sepsis in young children. World health organisation reported high burden of new born death caused by this bacterium. *Streptococcus agalactiae* colonizing epithelial cells of vagina and endothelial cells have high resistance to available antibiotic drugs which makes it essential to determine new drug targets.
Objectives: To compare the genome of selected strain with the non-pathogenic strains of streptococcus and identify the virulent and antibiotic resistant genes for adaptation in host environment.
Method: The whole genome of human pathogen *Streptococcus agalactiae* strain 2603 V/R was analysed and compared with *Streptococcus dysgalactiae* strains using visualization and annotation tools. Genomic islands, mobile genetic elements, virulent and resistant genes were studied.
Results: Genetically pathogenic strain is most similar to *Streptococcus dysgalactiae* subsp. *equisimilis* strain NCTC 7136. Comparative analysis revealed the importance of capsular polysaccharides and surface proteins responsible for avoiding immune system attachment to host epithelial cells and virulent behaviour. High number of genes coding for antibiotics resistance may provide a competitive advantage for survival of pathogenic *Streptococcus agalactiae* strain 2603 V/R in its niche.
Conclusions: The comparative analysis of pathogenic strain *Streptococcus agalactiae* with non-pathogenic strains of *Streptococcus dysgalactiae* provided new insights in pathogenicity that could aid in recognization for new regions and genes for development of new drug development strategies considering presence of high number of resistance genes.

1. Introduction

Streptococcus agalactiae of Streptococcaceae family is a spherical, gram-positive and Group B *Streptococcus* (GBS) bacterium, responsible for causing meningitis and sepsis in young children, mastitis in cattle and meningoencephalitis in fish (Gori et al., 2020). This bacterium is commonly associated with maternal infection, still birth, neonatal meningitis and infection in immune-compromised elderly human (Radtke et al., 2010; Furfaro et al., 2018). Within 28 days after birth, new-born meningitis inflames the meninges, the subarachnoid space, and the brain's blood vessels, which is associated with considerable mortality and long-term impairment. According to review, GBS prevalence in pregnant women in Brazil ranged from 4.2 to 28.4%, from last 10 years (et al., 2019). A WHO report concluded that global burden of GBS lead to 91,000 new born deaths, 46,000+ still births and 40,000 new-born living with neurological disability post infection in year 2020

Abbreviations: GBS, Group B *Streptococcus*; BBB, Blood Brain Barrier; MLST, Multilocus Sequence Typing; SDSE, *Streptococcus dysgalactiae subspecies equisimilis*; STSS, Streptococcal Toxic Shock Syndrome; CPS, Capsular polysaccharides; NPGP, NCBI Prokaryotic Genome Pipeline; RAST, Rapid Annotation Subsystem Technology server; IMG/M, Integrated Microbial Genomes and Microbiomes; CGView, Circular Genome Viewer; IS, Insertion Sequence; ANI, Average Nucleotide Identity; COG, Clusters of Orthologous Groups; GI, Genomic Island.

* Corresponding authors at: Maharshi Dayanand University, Rohtak 124001, Haryana, India.
E-mail addresses: mehak.bioinfo@mdurohtak.ac.in (M. Dangi), dir.research@mdurohtak.ac.in (A.K. Chhillar).
[1] Both authors contributed equally.

https://doi.org/10.1016/j.meegid.2022.105398
Received 3 November 2022; Received in revised form 19 December 2022; Accepted 22 December 2022
Available online 24 December 2022

(*Urgent Need for Vaccine to Prevent Deadly Group B streptococcus*, 2021). *Streptococcus agalactiae* is capable in forming attachment to numerous host cell types, specifically lung epithelial cells, vaginal epithelial cells and blood-brain barrier (BBB) micro-vascular endothelial cells (Silvestre et al., 2020).*Streptococcus agalactiae 2603 V/R*, a human pathogen, colonizes the gastrointestinal and genitourinary tracts (Silvestre et al., 2020). It can attach and colonize by adhesion to host cell/surface proteins and cause infection due to its ability to cross host tissue barriers and attach to epithelial cells of lung and vagina, avoiding host immune system and expressing virulent factors (Silvestre et al., 2020).The serotype of *Streptococcus agalactiae 2603 V/R* is determined by variations in the organisation of the capsular polysaccharides into distinctive repeating units (Radtke et al., 2010). *Streptococcus agalactiae* is identified by sequence type and a variety of clonal complexes when it is screened from human and animal sources. In recent years, multilocus sequence typing (MLST) has recognized six clonal complexes (CC1, CC10, CC17, CC19, CC23 and CC26) in humans, of which CC17 is characterised as hypervirulent and associated with serotype 3. In accordance with the location and distribution of clonal complexes characterizing genes in GBS genome, it is hypothesized that loss of function redundant genes evolved *Streptococcus agalactiae* in animals to evade human defence mechanism (Gori et al., 2020). This bacterium has resistance determinants to multiple antimicrobial drugs including beta-lactams, macrolids, licosamides, therefore, Penicillin-based therapy is used to treat GBS infection (Stewart et al., 2020).Comparative genome analysis helps to swiftly recognize gene mutations or deletions, dispensable genes and different proteins between strains with different virulence or host specificity to study pathogenicity and antigenic determinants (Wang et al., 2018).Bioinformatics has transformed clinical genetics making it possible to analyse hundreds of genes at an unprecedented speed. We have genomics pipelines to investigate susceptibility in whole genome and exome sequenced data for variant discovery, annotations, predictions, genotyping, and to compare patterns of genomic erosion (Ahmed and Zeeshan, 2021; Kutschera et al., 2022).

In this study, we didcomparative genomic analysis of pathogenic *Streptococcus agalactiae* strain *2603 V/R* with *Streptococcus dysgalactiae* strains in order to identify the antigenic and virulence associated genes that aids in development of neonatal meningitis.

The genome of pathogenic *Streptococcus agalactiae* strain *2603 V/R* used in our study has serotype V, for which development of vaccines is under clinical trial as per WHO factsheet. Identifying these unique and virulent genes might aid in development of control strategies by acting as drug targets as high antimicrobial resistance is observed in this pathogen (WHO, 2021).

Streptococcus dysgalactiae has subspecies named *Streptococcus dysgalactiae* subsp. *equisimilis* (SDSE) (Alves-Barroco et al., 2021).*Streptococcus dysgalactiae subsp. equisimilis* is increasingly evolving as a bacteria causing severe soft-tissue infections and acute systemic infections comprising (Streptococcal toxic shock syndrome) STSS (Rantala, 2014). SDSE usually infect horses and found in their respiratory tract causing contagious equine strangles disease and recently SDSE is emerging as a human pathogen causing infection of throat, skin and soft tissues (Baracco, 2019).

2. Materials and methodology

2.1. Strains selection

Nucleotide sequence (NC_004116.1) of assembled genome (ASM726v1) of pathogenic *Streptococcus agalactiae* strain 2603 V/R strain obtained from NCBI database with annotation done by NCBI prokaryotic genome pipeline (NPGP). *Streptococcus agalactiae* strain 2603 V/R was used as reference for BLAST standalone (Torkian et al., 2020) similarity search against non-pathogenic species using a customized database of available assembled complete genomes of *Streptococcus dysgalactiae* bacteria available on NCBI. Pathogenic strain is similar to *Streptococcus dysgalactiae* subsp. *equisimilis* strains ATCC 12394 and NCTC 7136 which we'll be considering non-pathogenic due to its inability to cause neonatal meningitis (Matsue et al., 2020).

2.2. Genome analysis

Functional Genome annotation of all three strains were performed using 1) RAST- Rapid Annotation Subsystem Technology server (Challa and Neelapu, 2019) which provides SEED-quality annotation including mapping of genes to subsystem. Assigned functions of the sequences by RAST were checked with BLAST (Challa and Neelapu, 2019) and IMG/M- Integrated Microbial Genomes and Microbiomes an online server (Michaud et al., 2008).

Genome visualization of *Streptococcus agalactiae* strain 2603 V/R was performed using.

1) Artemis- Annotation tool that was used for visualization of sequence feature (Carver et al., 2005).

2) CGView- Circular Genome Viewer for generating graphical maps for genome visualization (Fig. 1) (Grant and Stothard, 2008).

2.3. Comparative genome analysis

To compare pathogenic strain *Streptococcus agalactiae* strain 2603 V/R with non-pathogenic *Streptococcus dysgalactiae* subsp. *equisimilis* strains ATCC 12394 and NCTC 7136 we considered Genomic Islands (GI) and Insertion sequences (IS) as important features. GI are regions acquired through horizontal gene transfer and plays an essential role in adaptation, evolution and diversification of bacterial genomes (da Silva Filho et al., 2018). Island Viewer is used for the analysis and prediction of (GI) genomic islands of the genomes of all three strains (Dhillon et al., 2013) Island Viewer tool combines sequence composition GIs prediction methods SIGI-HMM (Langille et al., 2008) and Island Path DIMOB (Dhillon et al., 2013) with Comparative GI prediction method Island Pickfor highest overall accuracy (da Silva Filho et al., 2018). IS finder server was used along with IMG/M server to identify transposons and Insertion sequence(IS) in *Streptococcus agalactiae* strain2603 V/R (Hua et al., 2021).

RAST, IMG/M and EDGAR were used for comparative screening of gene content and presence of relevant coding sequence in genome of pathogenic and non-pathogenic strains. There results were independently checked by using BLAST nucleotide and protein queries in the genome. The phylogenetic profiler tool, part of IMG/M system at similarity cut-off of 60% was used for core genome estimation. Genes unique to *Streptococcus agalactiae* strain 2603 V/R were investigated using EDGAR (Dieckmann et al., 2021).

Virulence gene content of pathogenic *Streptococcus agalactiae* strain 2603 V/R was profiled using the database of virulence factors of pathogenic bacteria (Liu et al., 2019) through the interface at PATRIC (Davis et al., 2016). Virulence factors and pathogenic genes identified using PATRIC were investigated using RAST and IMG/M server to check their homologs in non-pathogenic strains and their results were confirmed by comparing it unique genes obtained through EDGAR.

3. Results and discussion

3.1. General features of pathogenic Streptococcus agalactiae strain 2603 V/R

Assembled genome available on NCBI database is 2.2 Mb sequence (ASM726v1) represented in 1 contig having 2,160,267 bp and GC content of the genome averages 35.6%. *Streptococcus agalactiae* 2603 V/R contains 1 chromosome in which there are 80 tRNA genes and 21 rRNA genes. There are 2124 protein coding genes (in IMG/M) which is more than protein coding genes present in *Streptococcus dysgalactiae* ATCC 12394 and NCTC 7136 strains.

The genome of pathogenic *Streptococcus* strain contains 1528

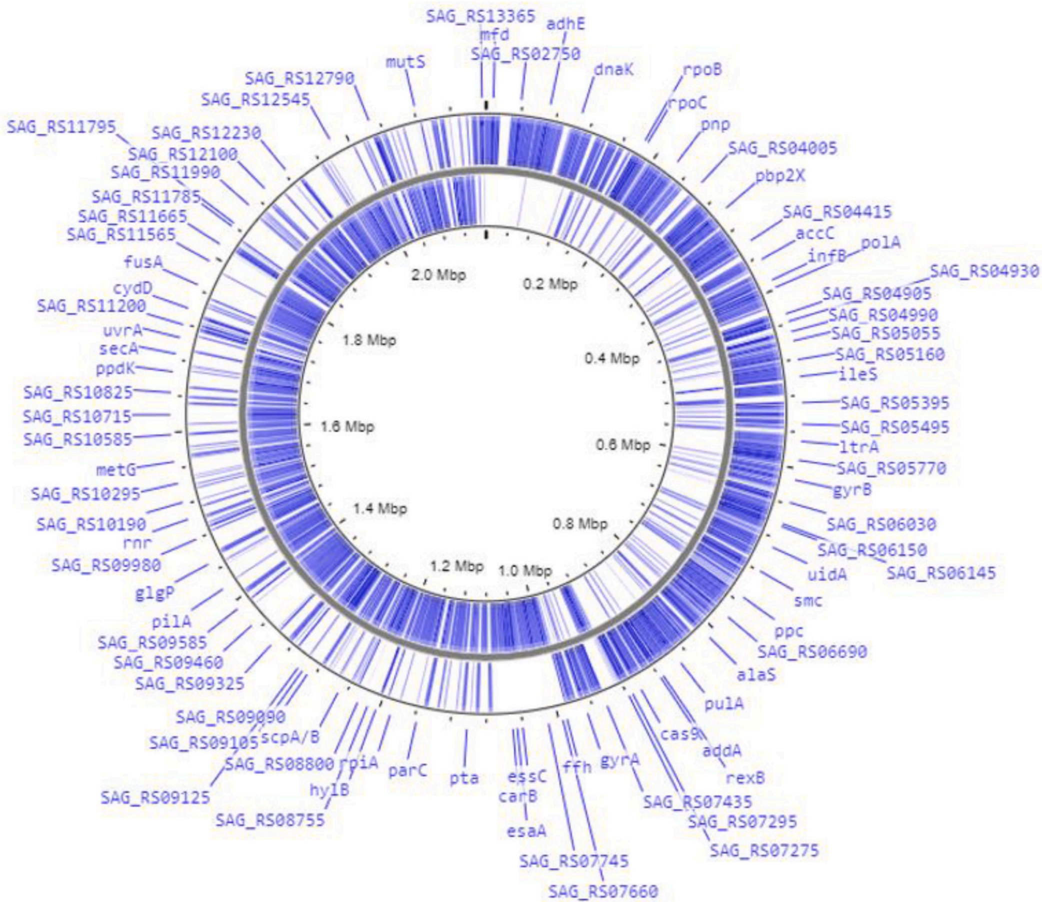


Fig. 1. Circular visualization of pathogenic strain Streptococcus agalactiae 2603 V/R using CG view tool.

(68.03%) protein coding genes with known function prediction and 596 (26.54%) protein coding genes which are annotated as either protein without known function or conserved hypothetical. Total number of genes detected in non-pathogenic NCTC 7136 strain is higher than in pathogenic strain 2603 V/R, whereas non-pathogenic strain ATCC 12394 has lower genes than both (Table 1).

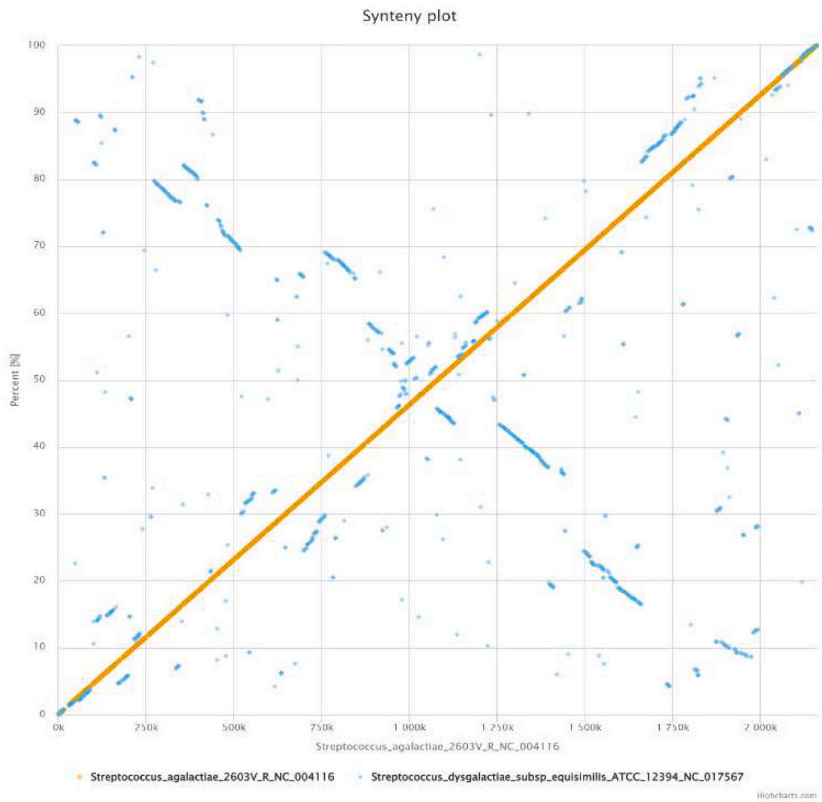
Number of genes coding signal peptide count is 118 (5.25%) in 2603 V/R. The average nucleotide identity (ANI) between genome of pathogenic *Streptococcus agalactiae* strain 2603 V/R and non-pathogenic strains was determined using ANIm application on Jspecies (Palmer et al., 2020). Resemblance of 98.12% is determined between 2603 V/R and ATCC 12394 whereas ANImresults for NCTC 7136 is 98.4%. To affirm conservation among the strains, the synteny plot demonstrated that on genomic scale comparison pathogenic *Streptococcus agalactiae* strain 2603 V/R is more similar to *Streptococcus dysgalactiae* strain NCTC 7136 than ATCC12394.

Table 1
Comparison of genomic characteristics (based on IMG/M server).

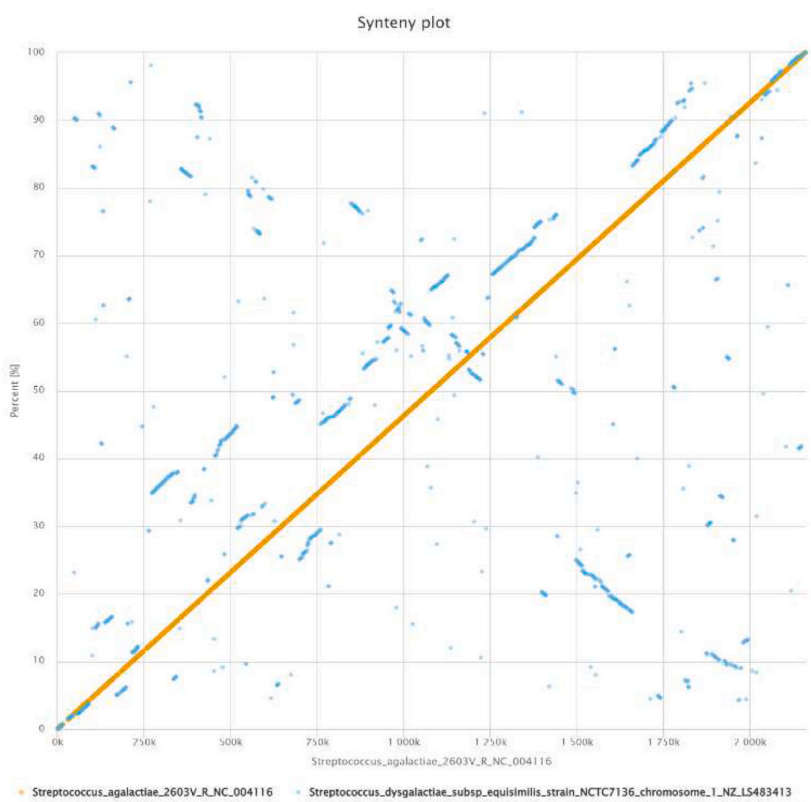
Genome	*Streptococcus agalactiae* 2603 V/R	*Streptococcus dysgalactiae subsp. equi* NCTC 7136	*Streptococcus dysgalactiae subsp. equi* ATCC 12394
Accession No.	NC_004116	NZ_LS483413	CP002215
Host	Human	Human	Human
Size	2,160,267	2,249,230	2,159,491
Genes(total)	2214	2257	2080
Protein coding genes	2124	2065	2056
CDS	2110	2168	2056
RNA	101	85	72
rRNA	7	6	15
tRNA	80	67	57
GC%	35.6%	39.4%	39.5%
Coding sequence	2155	2281	2227
No. of contigs	1	1	1
No. of subsystem	234	227	222

There are 611(28.7%) genes present in pathogenicstrain2603 V/R that are not detected in *Streptococcus dysgalactiae* NCTC7136. 14 of them are detected in *Streptococcus dysgalactiae* ATCC 12394 genome (Fig. 2). 597 are unique genes to 2603 V/R pathogenic strain found in chromosome, few of the genes belonged to hypothetical proteins (167). 34 of the specific gene to 2603 V/R are coding for ABC transporters, 15 are coding for acetyltransferases and only 2 genes SAG_RS07350 and SAG_RS07355 for tetracycline resistance and several other enzymes and proteins. 1313 genes are common in the genome of all three strains of *Streptococcus*. *Streptococcus agalactiae* 2603 V/R contains 15 genomic regions with an unusual GC content (Additional Table 2). Many of the genes with in these regions are associated with hypothetical protein, some contributed to the fitness of the strain (SAG_2025 coding for mn2+/fn2+ transporter, SAG_0924 coding for tetracycline resistance determinant leader peptide, SAG_2024 coding for mercuric resistance operon regulatory protein MerR) and some genes are associated with prophage lambda Sa2. Phage genes are associated with virulence and fitness of bacteria enhancing pathogenicity (Neely et al., 2020). Prophage lambda Sa2 (SAG1877, SAG1855) and Sa1 (SAG0548, SAG0555) phage anti-repressor proteins are in high number in pathogenic strain 2603 V/R than in non-pathogenic *Streptococcus dysgalactiae* strains. (See Fig. 3)

2603 V/R genome contains 21 genes coding for transposases which is



(A)



(B)

Fig. 2. Syntenic dotplots demonstrating the similarity of the genomes included in the analysis. A) S. aglactiae 2603 V/R (x axis) and S. dysgalactiae subsp. equi ATCC12394 B) S. agalactiae 2603 V/R and S. dysgalactiae subsp. equi NCTC 71361065(51.8%) homologousgenes to 2603 V/R are present in ATCC 12394 which are calculated using tool genome gene best homologs on IMG/M server with default parameter of 60+ percent identity.

Table 2
Mobile genetic elements present in the genome of Streptococcus agalactiae 2603 V/R (Based on the annotation results from RAST, IMG/M, and ED-GAR.

Locus Tag	Position	Length	Gene product	COG	Pfam
SAG2022	1,997,467–1,998,720	1254	Transposase, ISL3 family	COG3464	Pfam01610
SAG1253	1,261,820–1,263,127	1308	Transposase, ISL3 family	COG3464	pfam01610
SAG2002	1,983,230–1,983,619	390	Transposase, IS4 family	COG3293	pfam13359
SAG1994	1,974,440–1,974,766	327	Hypothetical protein pi2 Prophage protein 07	COG4707	pfam05595
SAG1877	1,862,833–1,863,543	711	Prophage LambdaSa2, antirepressor protein	COG3617	pfam02498
SAG1855	1,849,791–1,851,503	1713	Prophage LambdaSa2, terminase large subunit	COG4626	pfam03354
SAG1854	1,848,640–1,849,782	1143	Hypothetical protein, Phage portal protein BeeE	COG4695	pfam04860
SAG1853	1,848,048–1,848,590	543	Prohead peptidase. MEROPS family U35, Phage head maturation protease	COG3740	pfam04586
SAG1844	1,841,344–1,844,079	2736	Hypothetical protein, Phage-related minor tail protein	COG5280	pfam10145
SAG1795	1,787,517–1,788,683	1167	Transposase, IS30 family	COG2826	pfam13936
SAG1619	1,620,893–1,622,026	1134	IS1548 transposase	COG5433	pfam01609
SAG1584	1,589,092–1,590,225	1134	IS1548 transposase, YbfD/YdcC associated with H repeats	COG5433	pfam13808
SAG1550	1,559,166–1,559,555	390	IS1381 transposase protein B	COG3293	pfam13359
SAG1527	1,539,918–1,540,751	834	IS861, transposase OrfB	COG2801	pfam00665
SAG1521	1,533,654–1,534,820	1167	Transposase, IS30 family	COG2826	pfam13936
SAG1457	1,473,647–1,474,036	390	IS1381 transposase protein B	COG3293	pfam13359
SAG1268	1,277,301–1,277,993	693	Phage repressor protein C, transcriptional regulator, containspeptisase s24 and Cro/C1-type HTH domains	COG2932	pfam00717
SAG1244	1,254,329–1,255,108	780	ISSdy1, transposase OrfB	COG2801	pfam13276
SAG1241	1,253,409–1,253,684	276	Transposase OrfA, IS3 family	COG2963	pfam01527
SAG1235	1,244,650–1,245,927	1278	GBSi1, group II intron, maturase, Retron-type reverse transcriptase	COG3344	pfam08388 pfam00078
SAG1229	1,238,402–1,239,181	780	ISSdy1, transposase OrfB	COG2801	pfam00665
SAG1068	1,077,349–1,078,182	834	IS861, transposase OrfB	COG2801	pfam13333
SAG0966	973,262–973,651	390	IS1381 transposase protein B	COG3293	pfam13359
SAG0945	952,293–953,426	1134	IS1548 transposase	COG5433	pfam 13,808
SAG0760	754,678–75,581	1134	IS1548 transposase	COG5433	pfam01609
SAG0693	683,390–684,523	1134	IS1548 transposase	COG5433	pfam13808
SAG0640	628,888–629,163	276	Transposase OrfA, IS3 family	COG2963	pfam01527
SAG0639	628,019–628,840	822	Transposase OrfB, IS3 family	COG2801	pfam13276
SAG0596	585,209–587,221	2013	Prophage LambdaSa1, pblA protein, internal deletion	COG5412	pfam01609
SAG0585	579,635–581,050	1416	Hypothetical protein, phage terminase like protein	COG4626	pfam03354
SAG0567	569,396–570,715	1320	Prophage LambdaSa1, retron type reverse transcriptase	COG3344	pfam08388
SAG0555	563,950–564,699	750	Prophage LambdaSa1, antirepressor	COG3561	pfam03374
SAG0548	560,893–561,690	798	Prophage LambdaSa1, repressor proteinC, contains Cro/C1-type HTH and peptisase s24 domains	COG2932	pfam01381
SAG0543	557,682–558,071	390	IS1381 transposase protein B	COG3293	pfam13359
SAG0448	463,371–464,546	1176	Transposase, IS256 family	COG3328	pfam00872
SAG0300	310,678–311,196	519	Hypothetical protein, Uncharacterized SPBc2 prophage-derivedYoqJprotein	COG4474	pfam06908
SAG0261	268,657-269,046	390	Transposase, IS4 family	COG3293	pfam13359
SAG0195	215,240–216,373	1134	IS1548 transposase	COG5433	pfam13808
SAG0165	189,187–189,642	456	Competence protein ComGF	COG4940	pfam15980

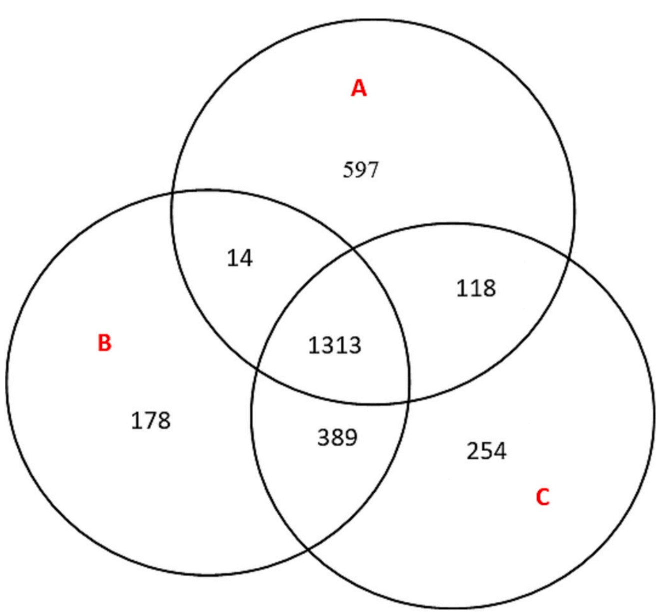


Fig. 3. Venn diagram dipicting distribution of protein coding genes among A) Streptococcus agalactiae 2603 V/R B) Streptococcus dysgalactiae subsp.equisimilis ATCC12394 C) Streptococcus dysgalactiae subsp. equisimilis NCTC7136.

much lower in comparison to 57 genes found in ATCC 12394. Detected insertion sequences in 2603 V/R genome belonged to ISSdy1, IS 861, ISL3, IS 4, IS 30, IS 1381and IS 861 families. A few genes have hypothetical protein as their product. ISSdy1 is a common IS element in *Streptococcus dysgalactiae* strain (Vasi et al., 2000). *Streptococcus dysgalactiae* NCTC 7136 commonly has IS3 family and has homologs in other streptococci including *Streptococcus agalactiae* and *Streptococcus pyogenes* as predicted by IS finder server. There is no pseudogene detected therefore these elements encode functional genes. The bacteria respond more adequately and swiftly to nutritious resources due to the presence of significant number of genes involved in ribosomal structure, translation and biogenesis helping during change of environment. Pathogenic strain 2603 V/R has more gene content in Cell wall/ envelope/ Capsule biogenesis category of clusters of orthologous groups (COG) than *Streptococcus dysgalactiae*atcc12394 which has more genes in defence mechanism COG category as detected used IMG/M.

3.2. Genomic Islands (GI)

The analysis of results of Island viewer server display that the *Streptococcus agalactiae* 2603 V/R have 21 regions with different GC content (Fig. 4). Genomic island as a result of horizontal gene transfer encodes antimicrobial resistance genes, virulence factors and bear metabolic pathways conferring specific adaptation (Bertelli et al., 2019). No 2603 V/R GI overlap with the 9 GI in NCTC 7136 and 11 GI in ATCC 12394. There are 12 genes in pathogen associated island (9676 bp)

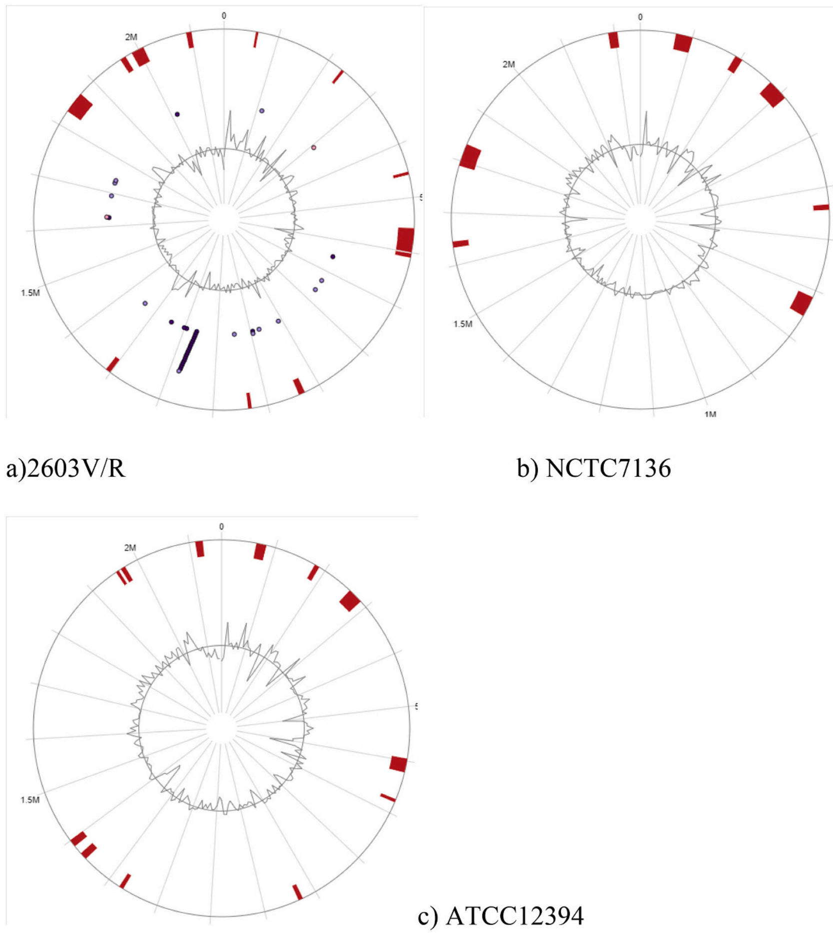


Fig. 4. Circular visualization of genomic islands and virulents factors described by islandviewer tool of A) Streptococcus agalactiae 2603 V/R B) Streptococcus dysgalactiae subsp. equisimilis NCTC7136 C) Streptococcus dysgalactiae subsp. equisimilis ATCC12394.

(SAG2110- SAG2121) which are absent in non-pathogenic *Streptococcus dysgalactiae* strains along with curated virulence factors such as streptolysin, capsules, botulinum toxin (additional Table 2). Most these genes encode product as hypothetical protein with 1 transcriptional regulator (SAG2118) and 1 recombinase (SAG2112). The pathogenic gene in this GI (SAG2121) is exclusively found in various *Streptococcus agalactiae* strains validated by blastp results and have hypothetical protein as product. The longest region found in *Streptococcus agalactiae* 2603 V/R is 40,560 in size and includes genes coding for prophage lambda Sa2, alkyl hyperoxide reductase, elongation factor Ts, 30S ribosomal proteinS2 and many hypothetical proteins.

Total size of these region accounts for 168 kb. SAG2024 have mercuric resistance operon regulatory protein as product in 19 genomic island is similar to lactobacillales protein (WP_000640387.1) upon blastx search. Genomic island 8 has gene SAG0924 tetracycline resistance determinant leader peptide as product which displays similarity to gene present in plasmid of *Enterococcus faecalis* strain FMAMOB6 and is also present in *Streptococcus pyogenes*. DNA damage inducible protein J of SAG0435 gene of genomic island 3 is also present in *Streptococcus dysgalactiae* and is a member of type 2 toxin-antitoxin system. Few genes in specific genomic islands encodes conjugated transposons (Tn916 and Tn 5252) and hypothetical proteins.Tn5252 is associated as a chromosome-borne element in the genome of *Streptococcus pneumonia (*Lannes-Costa et al., 2020*)*.

3.3. Pathogenicity and virulence

Pathogenic *Streptococcus agalactiae* 2603 V/R causes meningitis disease unlike *Streptococcus dysgalactiae sub equisimilis* strain NCTC7136 and ATCC12394 due to presence of i) virulent cell capsule and wall carbohydrate producing genes ii) Different surface proteins and enzymes that aids various process enhancing pathogenicity of Bacteria such as biofilm production and β-Hemolysin/cytolysin (β-HC, also *CylE*) toxin production iii) Genes for adherence to mucosal epithelial iv) Tansporters and gene regulators.

Streptococcus agalactiae strain 2603 V/R has type specific capsular polysaccharide (SAG1162-SAG1175) (Table 3). All polysaccharide genes involved in virulence behaviour are identified from genome of *Streptococcus agalactiae* 2603 V/R using BV-BRC-beta, RAST, IMG/M and are absent in *Streptococcus dysgalactiae* ATCC 12394 and NCTC7136. Capsular polysaccharides (CPS) are important for subdivision of species into 10 serotypes depending upon antigenic differences and are considered as key virulence factor (Stewart et al., 2020) and as expected results confirm no presence of CPS gene cluster in *Streptococcus dysgalactiae* NCTC7136 and *Streptococcus dysgalactiae* ATCC12394. Capsule polysaccharides A to J (Table 3)are essential for survival of bacteria in human blood as they act as barrier against immune system and absence of this cluster of genes in *Streptococcus dysgalactiae* makes it prone to human defence system (Noble et al., 2021). GroupStudies also prove that *CpsE* (SAG1171) is involved in synthesis of biofilm production (Chen et al., 2016) which provides protection from stress factors and enhance virulence and pathogencity (Silvestre et al., 2020).A study demonstrated that due to absence of *CpsE*, GBS are unable to evade phagocytosis by macrophages of placenta (Noble et al., 2021) which supports our result of inability of *Streptococcus dysgalactiae* strains to cause neonatal meningitis as no protein homolog of *CpsE* is found in

Table 3
Table presenting genes related with pathogenicity (based on results of PATRIC: BV-BRC server).

Sno.	Locus tag	Gene name	Product
1	SAG0737	*lgt*	Prolipoprotein diacyl glyceryl transferase
2	SAG1175	*cpsA*	Exopolysaccharide biosynthesis transcriptional activator EpsA Antiphagocytosis; Serum resistance
3	SAG1174	*cpsB*	Antiphagocytosis; Serum resistance, Tyrosine-protein phosphatase CpsB (EC 3.1.3.48)
4	SAG1173	*cpsC*	Tyrosine-protein kinase transmembrane modulator Antiphagocytosis; Serum resistance EpsC
5	SAG1172	*cpsD*	Tyrosine-protein kinase EpsD (EC2.7.10.2) Antiphagocytosis; Serum resistance
6	SAG1171	*cpsE*	Galactosyl transferase CpsE (EC 2.7.8.-) Antiphagocytosis; Serum resistance
7	SAG1170	*cpsF*	Polysaccharide biosynthesis protein CpsFAntiphagocytosis; Serum resistance
8	SAG1169	*cpsG*	hypothetical protein Antiphagocytosis; Serum resistance
9	SAG1168	*cpsH*	hypothetical protein Antiphagocytosis; Serum resistance
10	SAG1164	*cpsJ*	hypothetical protein Antiphagocytosis; Serum resistance
11	SAG1162	*cpsL*	Capsular polysaccharide repeat unit transporter Antiphagocytosis; Serum resistance
12	SAG1167	*cpsM*	Polysaccharide biosynthesis protein CpsM(V) Antiphagocytosis; Serum resistance
13	SAG1166	*cpsN*	hypothetical protein Antiphagocytosis; Serum resistance
14	SAG1165	*cpsO*	Glycosyl transferase, group 2 family Antiphagocytosis; Serum resistance
15	SAG1159	*neuD*	Sialic acid biosynthesis protein NeuD, *O*-acetyltransferase Antiphagocytosis; Serum resistance
16	SAG1160	*neuC*	UDP-*N*-acetyl glucosamine2-epimerase (EC 5.1.3.14) Antiphagocytosis; Serum resistance
18	SAG1161	*neuB*	*N*-acetyl neuraminate synthase (EC 2.5.1.56) Antiphagocytosis; Serum resistance
19	SAG1158	*neuA*	N-acylneuraminate cytidylyltransferase (EC 2.7.7.43) Antiphagocytosis; Serum resistance
20	SAG0406	*galU*	UTP-glucose-1-phosphate uridylyl transferase (EC 2.7.7.9) Antiphagocytosis; Adherence;Tissue invasion
21	SAG0667	*cylA*	ABC transporter, ATP-binding protein
22	SAG0668	*cylB*	Efflux ABC transporter, permease protein
23	SAG0663	*cylD*	hypothetical protein
24	SAG0669	*cylE*	hypothetical protein
25	SAG0670	*cylF*	hypothetical protein
26	SAG0664	*cylG*	Oxidoreductase, short-chain dehydrogenase/ reductase family
27	SAG0671	*cylI*	Polyketide synthase modules and related proteins
28	SAG0672	*cylJ*	hypothetical protein
29	SAG0673	*cylK*	hypothetical protein
30	SAG2043	*cfb*	CAMP factor Toxin; Membrane-damaging; Pore-forming
31	SAG1197		Hyaluronate lyase precursor (EC 4.2.2.1) Exoenzyme; Spreading factor
32	SAG1234	*lmb*	Laminin-binding surface protein Adherence
33	SAG1691	*scrB*	6-phosphate sucrose- hydrolase (EC 3.2.1.26)
34	SAG0967	*guaA*	GMP synthase [glutamine-hydrolyzing] (EC 6.3.5.2)/ amidotransferase subunit (EC 6.3.5.2), GMP synthase [glutamine-hydrolyzing]
35	SAG1742	*cydA*	Cytochrome-d-ubiquinol oxidase subunit I (EC 1.10.3.-)
36	SAG0305	*luxS*	Autoinducer-2 production protein LuxS @ S-ribosylhomocysteine lyase (EC 4.4.1.21)
37	SAG0028	*purN*	Phosphoribosyl glycinamide formyl transferase (EC 2.1.2.2)
38	SAG0047	*purB*	SAICAR lyase (EC 4.3.2.2) @ Adenylosuccinate lyase (EC 4.3.2.2)
39	SAG0030	*purH*	Phosphoribosyl aminoimidazole carboxamide formyl transferase (EC 2.1.2.3)/IMP cyclohydrolase (EC 3.5.4.10)
40	SAG0127	*fba*	Fructose-bisphosphate aldolase class II (EC 4.1.2.13)
41	SAG0450	*asnA*	Aspartate–ammonia ligase (EC 6.3.1.1)

Table 3 (*continued*)

Sno.	Locus tag	Gene name	Product
42	SAG1152	*ilvE*	(Branched chain) amino acid aminotransferase (EC 2.6.1.42)
43	SAG2069	*deoB-2*	Phosphopentomutase (EC 5.4.2.7)
44	SAG0298	*pbp1A*	Penicillin-binding protein 1A/1B (PBP1) / Multimodular transpeptidase-transglycosylase (EC 2.4.1.129) (EC 3.4.-.)
45	SAG1624	*csrS*	Transmembrane histidine kinase CsrS
46	SAG0788	*sodA*	Superoxide dismutase [Mn] (EC 1.15.1.1)
47	SAG2057	*leuS*	Leucyl-tRNA synthetase (EC 6.1.1.4)
48	SAG1086	*xpt*	Xanthine phosphor ribosyl transferase (EC 2.4.2.22)
49	SAG0906	*lepA*	Translation elongation factor LepA
50	SAG1763	*glnA*	type I Glutamine synthetase (EC 6.3.1.2)
51	SAG1625	*csrR*	Response regulator CsrR
52	SAG1585	*clpP*	protease proteolytic subunit ClpP ATP-dependent Clp (EC 3.4.21.92)
53	SAG0525	*aspC*	Aspartate amino transferase (EC 2.6.1.1)
54	SAG0969	*gid*	Methylene tetrahydrofolate–tRNA-(uracil-5-)-methyl transferase TrmFO (EC 2.1.1.74)
55	SAG0768	*murF*	UDP-*N*-acetyl muramoyl-tripeptide–D-alanyl-D-alanine ligase (EC 6.3.2.10)
56	SAG0107	*pyrG*	CTP synthase (EC 6.3.4.2)
57	SAG0986	*pepN*	Lysyl amino peptidase (EC 3.4.11.15)
58	SAG1042	*carB*	Carbamoyl-phosphate synthase large chain (EC 6.3.5.5)
59	SAG1182	*deoB-1*	Phosphopentomutase (EC 5.4.2.7)
60	SAG0297	*pepC*	Amino peptidase C (EC 3.4.22.40)
61	SAG1736	*pepX*	Xaa-Pro-dipeptidyl-peptidase (EC 3.4.14.11)
62	SAG0985	*ciaR*	Two component system response regulator CiaR
63	SAG0988		ATP-binding protein PstB (TC 3.A.1.7.1), Phosphate ABC transporter
64	SAG0707		Catabolite control protein A
65	SAG1462		Cell-wall-anchored protein SasA (LPXTG motif)
66	SAG0990		permease protein PstA (TC 3.A.1.7.1), Phosphate ABC transporter
67	SAG1176		transport regulator MtaR, LysR family and Methionine biosynthesis
68	SAG0991		permease protein PstC (TC 3.A.1.7.1), Phosphate ABC transporter
69	SAG1761		Ribonuclease J1 (endonuclease and 5′ exonuclease)
70	SAG0720		Two-component sensor kinase SA14–24
71	SAG1531		Inner membrane permease protein SitD, Manganese ABC transporter
72	SAG0106		RNA polymerase delta subunit (DNA-directed) (EC 2.7.7.6)
73	SAG0636		DUF1706 domain-containing protein
74	SAG0025		Phosphoribosylformylglycinamidine synthase, glutamine amidotransferase subunit (EC 6.3.5.3) / Phosphoribosylformylglycinamidine synthase, synthetase subunit (EC 6.3.5.3)
75	SAG1332		Transcriptional regulator, MarR family
76	SAG1904		2-deoxy-D-gluconate 3-dehydrogenase (EC 1.1.1.125) @ 2-dehydro-3-deoxy-D-gluconate 5-dehydrogenase (EC 1.1.1.127)
77			Nicotinamide-nucleotide amidase (EC 3.5.1.42) / ADP-ribose pyrophosphatase of COG1058 family (EC 3.6.1.13)
78	SAG1434		Rhodanese domain protein UPF0176, Firmicutes subgroup
79	SAG0666		FabZ form (EC 4.2.1.59), 3-hydroxyacyl-[acyl-carrier-protein] dehydratase
80	SAG0662		hypothetical protein
81			C5a peptidase (EC 3.4.21.110) Immune evasion; Complement protease; Adherence;Surface-associated adhesin
82	SAG0645		Cell wall surface anchor family protein Adherence; pili assembled Sortase
83	SAG0649		hypothetical protein Adherence; pili assembled Sortase
84	SAG1408		hypothetical protein Adherence; pili assembled Sortase
85	SAG1052		hypothetical protein Adherence;Phase variation
86			Streptococcal extracellular nuclease 3; Mitogenic factor 3 Exoenzyme; Spreading factor
87	SAG1404		hypothetical protein Adherence;Sortase-assembled pili

(*continued on next page*)

Table 3 (*continued*)

Sno.	Locus tag	Gene name	Product
88	SAG0665		Acyl carrier protein
89	SAG1405		Sortase A, LPXTG specific Adherence;Sortase-assembled pili
90	SAG0647		Sortase A, LPXTG specific Adherence;Sortase-assembled pili
91	SAG0648		Sortase A, LPXTG specific Adherence;Sortase-assembled pili

their genome. Thus, *cpsE* proves to be a chief virulence factor in causing infection in reproductive tract of pregnant woman and foetus. Capsule anchors surface associated proteins that help in immune inactivation for example C5a peptidase, an enzyme specific to pathogenic strain is undetected in non-pathogenic strains of *Streptococcus dysgalactiae*.

Product of *NeuD* gene, sialic acid is attached to CPS of *Streptococcus agalactiae* interfere with complement mediated killing by human immune cells which has no homologous protein in *Streptococcus dysgalactiae* strains. This results in immune action by host interfering with development of infection in case of *Streptococcus dysgalactiae (*Alhhazmi and Tyrrell, 2018*)*. Gene *neuD* absent in *Streptococcus dysgalactiae* strains also provide resistance to antimicrobial activities of platelets by blocking platelet activation (Uchiyama et al., 2019).This works as additional factors to support the thesis that due to absence of cps genes and associated proteins, *Streptococcus dysgalactiae* strains NCTC7136 and ATCC12394 might use different mechanism for survival in human and are unable to cause serious infection.

Few virulent genes of *Streptococcus agalactiae*strain2603 V/R encode hypothetical proteins (Table 3) as displayed by BV-BRC PATRIC server. Gene coding for a transcriptional regulator, Arc family with locus ID SAG0644 is absent in both non-pathogenic strains. According to RAST results it has possible role in multiple sugar metabolism responsible for adhesion and belongs to *Streptococcus pygones* recombinational zone. On performing BLASTp against *Streptococcus dysgalactiae* ATCC12394 for this gene SAG0644, presence of function homolog with percent identity 26.96 is recognized. A study confirms *Streptococcus pygones* recombinational zone involvement in adaptation to host environment during infection (Bessen and Kalia, 2002).

Cfb gene encoding CAMP factor providing defence against immune system and can damage host cells (Palang et al., 2020), is a marker for GBS strains and reveal no protein homologs in *Streptococcus dysgalactiae* strains.

There are some common surface proteins present in all strains for example Laminin-binding surface protein which binds to zinc with high affinity (Sridharan et al., 2019) and is essential for bacterial grow this one such common protein responsible for adherence (Moulin et al., 2016). We identified some proteins and enzymes (SNo. 31–62(Table 3)) related to virulence that are mutual in both species but have less percent similarity to original protein symbolising presence of mutation in genomic sequence of non-pathogenic *Streptococcus dysgalactiae* strains.

csrS gene (SAG1624) has 50% similar amino acid similarity in *Streptococcus dysgalactiae* strain where it's regulator (SAG1625) is 84% similar and has both repressor and activator activity. Mutation in *csrS* gene results in lesser ability to invade microvascular endothelial cells of brain (Thomas et al., 2020), therefore lesser similarity of protein sequence in *Streptococcus dysgalactiae* strains might be the reason for decrease in effectiveness of this gene leading to downregulation in its function and inability to cause neonatal meningitis. Another study supports loss in function mutation in *csrRS* gene affects its repressor activity leads to increase in virulent behaviour of the bacterial strains (Lynskey et al., 2019).

RocA gene responsible for regulation of *csrRS* gene which represses gene expression (Lynskey et al., 2019), is present in *Streptococcus dysgalactiae* strain ACTC12394 and not in *Streptococcus agalactiae* strain therefore result in decreased action of virulent factors and genes that are present in non-pathogenic strains and hence responsible for not causing meningitis or severe infection.

The *cylE* (Sag0669) gene encode surface protein hemolysin/cytolysin (ßhc) responsible for pore formation in *Streptococcus agalactiae*-strain2603 V/R (Tettelin et al., 2002) has no protein homolog in *Streptococcus dysgalactiae* strains. ßhc toxin is responsible for decreased barrier resistance at maternal-fetal interface, lungs and brain as well as, is cytotoxic to human immune system cells therefore, it is considered to be major reason for pathogenicity.

PbsP plasminogen binding surface is conserved among GBS strains and its presence promotes bacterial adherence and epithelial cell invasion in 2603 V/R, it belongs to pfam protein family HMM PF00746 and similar protein sequence is not detected *in Streptococcus dysgalactiae* strains upon using selected genome blastp search on IMG/M pipeline along with another cell wall anchored protein SasA (LPXTG motif) at locus ID SAG 1462. Absence of PbsP protein creates hinderance in interaction with host vitonectin (acts as a bridge between SSURE domain of PbsP and host integrins) (De Gaetano et al., 2018).

Above observations correlate with assumptions that pathogenic *Streptococcus agalactiae* 2603 V/R contains virulence determinants responsible for meningitis in human that are not present in *Streptococcus dysgalactiae* strains determined by comparative genome analysis. Some virulence factors are also present in *Streptococcus dysgalactiae* ATCC12394 and NCTC 7136 indicating their inessential or altered function due to mutation to minimize the pathogenicity. A large number of genes that are related with ABC transport and cylA that are unique to pathogenic strain and responsible for defence mechanism. Few virulent genes (Table 3) that are related to pathogenicity as demonstrated by result of BV-BRC server encode hypothetical proteins.

3.4. Comparison with E. coli and Klebsiella

Neonatal meningitis disease is classified as early or late onset. GBS even after presence of intrapartum antibiotics remains as the most common cause (40% to 50%). Next to GBS, *E. coli* is the second most common pathogen followed by *Klebsiella.* Also the Gram negative bacilli such as *Klebsiella and E. coli* may be more common than GBS in developing countries (Khalessi and Afsharkhas, 2014).

While comparing virulent genes with *E. coli* and *Klebsiella pneumoniae* it was observed that 45 virulent genes have no homologs in *E. coli* isolate K1 and *Klebsiella pneumoniae.* Other genes even though have protein homologs present in *E. coli* and *Klebsiella pneumoniae, but it was observed that* percent identity in such cases is less than 50% with only very few exceptions. Both *E. coli* isolate K1 and *Klebsiella pneumoniae* have 37 protein homologs of virulent genes present in *Streptococcus agalactiae* 2603 V/R, while 4 more homologs are specifically present in *E. coli. Klebsiella pneumoniae* isolate K1 only contains 2 such specific homolog in its genome.(Table 4)

3.5. Antibiotic resistance

In the genome of *Streptococcus agalactiae*2603 V/R several additional genes that are responsible for antimicrobial resistance were detected (Table 5). They code for L-O-lysyl phosphatidyl glycerol synthase (SAG2131), Aminoglycoside 6-nucleotidyltransferase (SAG 2029) for resistance to aminoglycosides. Tetracycline resistance protein TetM (SAG0923)display highest similarity to protein found in *mycoplasmmycoidescapri* GM12, *E. coli*, *Clostridioides difficile* and *salmonella entrica* along with protein homologs in various *Streptococcus agalactiae* strains whereas genetic sequence similarity of gene upon blastn resembles most to *E. faecalis*, *Streptococcus gallolyticus*, *Streptococcus canis* and *E. coli* recommending origin from these bacteria. In addition, the above-mentioned resistance genes, genome of *Streptococcus agalactiae*-strain2603 V/R contains drug efflux transporters SAG_RS08840 MATE family efflux transporter and SAG_RS11120 multidrug efflux MFS transporter.

Table 4
Comparison of identified virulent gene protein homologs present in *Streptococcus agalactiae* 2603 V/R with *E. coli* K1 and *Klebsiella pneumoniae* K. (results were based on analysis using IMG/M server).

Sno.	Streptococcus Agalactiae 2603 V/R	Gene name	*E. coli* isolate K1 (Gu Hao et al., 2018) (Percent identity)	*Klebsiella pneumoniae* isolate K1 (Carrie et al., 2019) (Percent identity)
1	SAG0737	*lgt*	No Homolog	No Homolog
2	SAG1175	*cpsA*	No Homolog	No Homolog
3	SAG1174	*cpsB*	No Homolog	No Homolog
4	SAG1173	*cpsC*	35.38	No Homolog
5	SAG1172	*cpsD*	33.01	38.09
6	SAG1171	*cpsE*	29.79	26.79
7	SAG1170	*cpsF*	No Homolog	No Homolog
8	SAG1169	*cpsG*	No Homolog	No Homolog
9	SAG1168	*cpsH*	No Homolog	No Homolog
10	SAG1164	*cpsJ*	46.32	32.31
11	SAG1162	*cpsL*	No Homolog	No Homolog
12	SAG1167	*cpsM*	40.00	No Homolog
13	SAG1166	*cpsN*	30.77	25.00
14	SAG1165	*cpsO*	36.04	35.97
15	SAG1159	*neuD*	33.75	41.18
16	SAG1160	*neuC*	20.62	No Homolog
18	SAG1161	*neuB*	No Homolog	No Homolog
19	SAG1158	*neuA*	26.1	41.4
20	SAG0406	*galU*	43.88	46.08
21	SAG0667	*cylA*	31.68	30.64
22	SAG0668	*cylB*	No Homolog	No Homolog
23	SAG0663	*cylD*	No Homolog	No Homolog
24	SAG0669	*cylE*	No Homolog	No Homolog
25	SAG0670	*cylF*	No Homolog	No Homolog
26	SAG0664	*cylG*	43.28	42.44
27	SAG0671	*cylI*	No Homolog	No Homolog
28	SAG0672	*cylJ*	No Homolog	No Homolog
29	SAG0673	*cylK*	No Homolog	No Homolog
30	SAG2043	*cfb*	No Homolog	No Homolog
31	SAG1197		No Homolog	No Homolog
32	SAG1234	*lmb*	24.71	27.40
33	SAG1691	*scrB*	35.43	30.47
34	SAG0967	*guaA*	52.98	52.40
35	SAG1742	*cydA*	34.84	32.23
36	SAG0305	*luxS*	No Homolog	No Homolog
37	SAG0028	*purN*	No Homolog	No Homolog
38	SAG0047	*purB*	No Homolog	No Homolog
39	SAG0030	*purH*	49.53	50.09
40	SAG0127	*fba*	No Homolog	43.00
41	SAG0450	*asnA*	62.42	63.64
42	SAG1152	*ilvE*	32.61	31.33
43	SAG2069	*deoB-2*	39.72	40.09
44	SAG0298	*pbp1A*	No Homolog	No Homolog
45	SAG1624	*csrS*	No Homolog	No Homolog
46	SAG0788	*sodA*	51.44	51.92
47	SAG2057	*leuS*	42.66	42.84
48	SAG1086	*xpt*	31.88	30.88
49	SAG0906	*lepA*	57.93	*58.60*
50	SAG1763	*glnA*	41.53	*40.22*
51	SAG1625	*csrR*	42.53	38.67
52	SAG1585	*clpP*	57.22	57.22
53	SAG0525	*aspC*	No Homolog	No Homolog
54	SAG0969	*gid*	40.48	39.58
55	SAG0768	*murF*	No Homolog	No Homolog
56	SAG0107	*pyrG*	50.09	51.20
57	SAG0986	*pepN*	24.83	25.50
58	SAG1042	*carB*	49.49	49.86
59	SAG1182	*deoB-1*	39.72	40.09
60	SAG0297	*pepC*	No Homolog	No Homolog
61	SAG1736	*pepX*	No Homolog	No Homolog
62	SAG0985	*ciaR*	No Homolog	No Homolog
63	SAG0988		51.01	50.61
64	SAG0707		No Homolog	No Homolog
65	SAG1462		No Homolog	No Homolog
66	SAG0990		No Homolog	No Homolog
67	SAG1176		23.34	19.36
68	SAG0991		No Homolog	No Homolog
69	SAG1761		No Homolog	No Homolog
70	SAG0720		No Homolog	No Homolog
71	SAG1531		31.42	32.95
72	SAG0106		No Homolog	No Homolog
73	SAG0636		No Homolog	No Homolog
74	SAG0025		25.46	24.87
75	SAG1332		24.31	25.00
76	SAG1904		40.00	40.78
77	SAG1434		29.34	31.77
78	SAG0666	*cylZ*	41.38	42.24
79	SAG0662		No Homolog	No Homolog
80	SAG0645		No Homolog	No Homolog
81	SAG0649		No Homolog	No Homolog
82	SAG1408		No Homolog	No Homolog
83	SAG1052		No Homolog	No Homolog
84	SAG1404		No Homolog	No Homolog
85	SAG0665		41.77	No Homolog
86	SAG1405		No Homolog	No Homolog
87	SAG0647		No Homolog	No Homolog
88	SAG0648		No Homolog	No Homolog
89	EC 3.4.21.110	C5a peptidase	No Homolog	No Homolog
90	EC 3.5.1.42	Nicotinamide-nucleotide amidase	No Homolog	Enzyme is Present

Table 5
Streptococcus agalactiae 2603 V/R genes associated with antimicrobial resistance (based on PATRIC).

RefSeq Locus Tag	Gene	Product
SAG0160	*rpoB*	RNA polymerase beta subunit (DNA-directed) (EC 2.7.7.6)
SAG0322		Cell envelope stress response system LiaFSR, response regulator LiaR(VraR)
SAG0057	*rpsJ*	SSU ribosomal protein S10p (S20e)
SAG1769	*fusA*	Translation elongation factor G
SAG2152	*pgsA*	CDP-diacylglycerol–glycerol-3-phosphate 3-phosphatidyl transferase (EC 2.7.8.5)
SAG0960	*gyrA*	(Subunit A) DNA gyrase (EC 5.99.1.3)
SAG2131		L-O-lysylphosphatidyl glycerol synthase (EC 2.3.2.3)
SAG1778		23S rRNA (guanine(748)-N(1))-methyltransferase (EC 2.1.1.188)
SAG0346	*fabK*	Enoyl-[acyl-carrier-protein] reductase [FMN, NADH] (EC 1.3.1.9), FabK ≥ refractory to triclosan
SAG0349	*fabF*	KASII (EC 2.3.1.179), 3-oxoacyl-[acyl-carrier-protein] synthase
SAG0320		specific inhibitor of LiaRS(VraRS) signaling pathway, Membrane protein LiaF(VraT)
SAG1115	*folP*	Dihydropteroate synthase (EC 2.5.1.15)
SAG0843	*murA-1*	UDP-*N*-acetyl glucosamine 1-carboxy vinyl transferase (EC 2.5.1.7)
SAG0485	*ileS*	Iso leucyl-tRNA synthetase (EC 6.1.1.5)
SAG1684	*Alr*	Alanineracemase (EC 5.1.1.1)
SAG0767	*Ddl*	D-alanine ligase–D-alanine (EC 6.3.2.4)
SAG2029		Aminoglycoside 6-nucleotidyltransferase, putative
SAG1771	*rpsL*	SSU ribosomal protein S12p (S23e)
SAG0321		sensor histidine kinase LiaS(VraS), Cell envelope stress response system LiaFSR
SAG0923	*tetM*	ribosomal protection type ≥ Tet(M), Tetracycline resistance
SAG0866	*murA-2*	UDP-*N*-acetyl glucosamine 1-carboxyvinyl transferase (EC 2.5.1.7)
SAG0073	*rplF*	L6p LSU ribosomal protein (L9e)
SAG0623	*gyrB*	(Subunit B) DNA gyrase (EC 5.99.1.3)
SAG0762	*Tuf*	Tu Translation elongation factor
SAG0161	*rpoC*	DNA-directed RNA polymerase beta' subunit (EC 2.7.7.6)
SAG1629	*gidB*	16S rRNA (guanine(527)-N(7))-methyl transferase (EC 2.1.1.170)
SAG1314	*folA*	Dihydro folate reductase (EC 1.5.1.3)

There is presence of considerable number of GNAT superfamily acetyltransferase suggesting possible action on aminoglycosides with antimicrobial characteristics. Various genes and proteins of *Streptococcus agalactiae*2603 V/R strain (Table 5) are responsible for providing resistance to large categories of antibiotics including Fosfomycin, cycloserine and fluroquinoloes suggesting use of penicillin as a reliable drug for treatment (Stewart et al., 2020). This ability of pathogenic *Streptococcus agalactiae* 2603 V/R to cope antibiotics action inside physiological environment of the host organism is added advantage for its survival during the infection. *Streptococcus agalactiae* 2603 V/R has a unique presence of mprF gene when compared to *Streptococcus dysgalactiae* strains, providing resistance to cationic peptides. *Streptococcus dysgalactiae* strains also encodes presence of some common antibiotic resistance genes (SAG_RS05125, SAG_RS09020, SAG_RS10350 genes of *Streptococcus agalactiae* 2603 V/R) ED-GAR and their homologs on islandviewer but are considerably lower in number implying better availability of treatment drugs.

3.6. Toxin-antitoxin system

There is presence of three toxin-anti toxin system genes in *Streptococcus agalactiae* strain 2603 V/R and are unique to it (Additional Table 2) (SAG_RS03850, SAG_RS05000, SAG_RS11755) raising questions regarding their origin and potential physiological function of bacterium. Difference in the antitoxin systems and detected toxin in *Streptococcus* species may be because of different niche and inhabited host and increases the fitness of this bacterium. Type II toxin-antitoxin have important roles in bacterial virulence, formation of biofilm, maintenance of genetic material and drug tolerance which are feature of *Streptococcus agalactiae*strain2603 V/R (Kamruzzaman et al., 2021).

1) RelE/ ParE family toxin SAG 0228 (type II toxin-antitoxin system): Blastp analysis of this gene revealed high similarity to proteins from various pathogenic *Streptococcus* species including *Streptococcus pyogenes* strains.
2) RelB/ DinJ family antitoxin SAG0435 gene that encodes a type II toxin-antitoxin system and DNA damage inducible protein J.
3) Toxin-antitoxin type II HicA family toxin annotated as hypothetical protein in img/m pipeline

The top homologous protein of above mentioned are present in various *Streptococcus agalactiae* strains. Pathogenic strain shares some antitoxin and toxin system with *Streptococcus dysgalactiae* strains that is Txe/YoeB family addiction module toxin, type II toxin-antitoxin system prevent-host-death family antitoxin, nucleotidyl transferase AbiEii/ AbiGii toxin family protein.

Table 6 summarises the above results and analysis data regarding presence of genes using Edgar and IMG/M server.

Table 6
Presence of genes studied using Edgar and IMG/M server.

Streptococcus agalactiae 2603 V/R	Total genes	2214
Virulent genes	Total virulent genes	91
	Unique *S. agalactiae 2603 V/R*	45
	Similar to *S. dysgalactiae ATCC12394*	19
	Similar to *S. dysgalactiae NCTC7136*	0
	Virulent genes common in all strains	23
Toxin-anti toxin system genes	Total toxin-antitoxin system genes	7
	Unique to *S. agalactiae 2603 V/R*	4
	Similar to *S. dysgalactiae ATCC12394*	0
	Similar to *S. dysgalactiae NCTC7136*	0
	Present in all strains	3
Resistance related gene	Total resistance related genes	27
	Unique *S. agalactiae 2603 V/R*	3
	Similar to *S. dysgalactiae ATCC12394*	4
	Similar to *S. dysgalactiae NCTC7136*	1
	Present in all strains	19

4. Conclusion

The genomic analysis of sequence of the pathogenic *Streptococcus agalactiae 2603 V/R* discovered that this strain was adapted to a pathogenic lifestyle reflecting lack of prominent virulence factors present in *Streptococcus dysgalactiae*that could not cause neonatal meningitis, and it was analysed that pathogenic strain was marked by the presence of genes encoding enzymes involved in adherence to host cell, in addition to genes responsible for defence from immune cells of host. *Streptococcus agalactiae* contained high number of resistance genes some of which are also present in *Streptococcus dysgalactiae* strains that helps in adapting to environment. Some of the virulence elements mostly enzymes are also present innon-neonatal meningitis Causing*streptococcus dysgalactiae* strains as demonstrated by our results, which implies that, these genes were rather contribute to general functions of the bacterium or aid in adaptive abilities in the same host to cause different or less severe infection by down regulation done by repressor gene. Inability of *Streptococcus dysgalactiae* strains to cause meningitis in human depends upon presence of additional virulent genes that were capsular polysaccharides (SAG1162-SAG1175) and allows host colonization. *Streptococcus agalactiae 2603 V/R* contains cps genes, responsible for capsular biogenesis and are virulent in behaviour which are absent in *Streptococcus dysgalactiae* along with some other important surface protein encoding generesponsible for attachment to host epithelial cells in lungs, vagina and brain. Dueto lack of possible horizontal gene transfer in *Streptococcus dysgalactiae*or loss in functioning of genes present in both the species also constituent the basis of pathogenicity of *Streptococcus agalactiae* 2603 V/R in humans to cause neonatal meningitis. Because of ability of *Streptococcus dysgalactiae* to invade human cells similar to *Streptococcus agalactiae* and yet not able to cause maternal and neonatal disease similar to *Streptococcus agalactiae* 2603 V/R irrespective of common lineage and horizontal genetic transfer from *Streptococcus pyogens*, it will be interesting toinvestigate more such similar strains to reveal the similar genetic features and proteins responsible or enhancing *Streptococcus agalactiae* pathogenicity.

At last, comparative analysis of *Streptococcus agalactiae* will allow generating more knowledge about these bacteria that could aid in recognizing genetic regions for development of identification primers for control strategies and proteins that might serve as possible targets for drugs considering presence of high number of resistance genes in these genomes.

Data statement

Datasets analysed during the current study are available on the national center for biotechnology (NCBI) database https://www.ncbi.nlm.nih.gov/nuccore. These datasets were derived from the following public domain resources:

1. Nucleotide sequence of *Streptococcus agalactiae* strain 2603 V/R: https://www.ncbi.nlm.nih.gov/nuccore/NC_004116.1
2. Nucleotide sequence of *Streptococcus dysgalactiae* subsp. *equisimilis* strain ATCC 12394: https://www.ncbi.nlm.nih.gov/nuccore/CP002215.1
3. Nucleotide sequence of *Streptococcus dysgalactiae* subsp. *equisimilis* strain NCTC 7136: https://www.ncbi.nlm.nih.gov/nuccore/1409013694

CRediT authorship contribution statement

Jasmine Arya: Methodology, Software, Writing – original draft, Writing – review & editing. **Divya Sharma:** Methodology, Software, Writing – original draft. **Dev Kumar:** Software, Resources. **Ritu Jakhar:** Resources, Data curation. **Alka Khichi:** Data curation. **Mehak Dangi:** Conceptualization, Supervision. **Anil Kumar Chhillar:** Supervision.

Data availability

I have shared link to the data at the attach file step

Acknowledgement

We want to acknowledge "DBT Builder Maharshi Dayanand University Interdisciplinary Life Science Programme for Advance Research and Education" for providing the necessary support and facilities for the completion of this research work.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.meegid.2022.105398.

References

Ahmed, Z., Zeeshan, S., 2021. Genomics pipelines to investigate susceptibility in whole genome and exome sequenced data for variant discovery, annotation, prediction and genotyping. PeerJ 9, e11724.
Alhhazmi, A., Tyrrell, G.J., 2018. Phenotypic and molecular analysis of nontypeable group B streptococci: identification of cps2a and hybrid cps2a/cps5 group B streptococcal capsule gene clusters. Emerg. Microbes Infect. 7 (1), 1–12.
Alves-Barroco, C., et al., 2021. New insights on Streptococcus dysgalactiae subsp. dysgalactiae isolates. Front. Microbiol. 12, 686413.
Baracco, G.J., 2019. Infections caused by group C and G streptococcus (Streptococcus dysgalactiae subsp. equisimilis and others): epidemiological and clinical aspects. Microbiol. Spectrum 7 (2) (p. 7.2. 32).
Bertelli, C., Tilley, K.E., Brinkman, F.S., 2019. Microbial genomic island discovery, visualization and analysis. Brief. Bioinform. 20 (5), 1685–1698.
Bessen, D.E., Kalia, A., 2002. Genomic localization of a T serotype locus to a recombinatorial zone encoding extracellular matrix-binding proteins in Streptococcus pyogenes. Infect. Immun. 70 (3), 1159–1167.
Carrie, C., Levy, C., Alexandre, C., Baleine, J., et al., 2019. Klebsiella pneumoniae and Klebsiella oxytoca meningitis in infants. Epidemiological and clinical features. Arch. Pediatr. 26 (1), 12–15.
Carver, T.J., et al., 2005. ACT: the Artemis comparison tool. Bioinformatics 21 (16), 3422–3423.
Challa, S., Neelapu, N.R.R., 2019. Phylogenetic trees: applications, construction, and assessment. Essentials Bioinforma. III, 167–192.
Chen, I.-M.A., et al., 2016. IMG/M: integrated genome and metagenome comparative data analysis system. Nucleic Acids Res. gkw929.
da Silva Filho, A.C., et al., 2018. Comparative analysis of genomic island prediction tools. Front. Genet. 9, 619.
Davis, J.J., et al., 2016. Antimicrobial resistance prediction in PATRIC and RAST. Sci. Rep. 6 (1), 1–12.
De Gaetano, G.V., et al., 2018. The Streptococcus agalactiae cell wall-anchored protein PbsP mediates adhesion to and invasion of epithelial cells by exploiting the host vitronectin/αv integrin axis. Mol. Microbiol. 110 (1), 82–94.
Dhillon, B.K., et al., 2013. IslandViewer update: improved genomic island discovery and visualization. Nucleic Acids Res. 41 (W1), W129–W132.
Dieckmann, M.A., et al., 2021. EDGAR3. 0: comparative genomics and phylogenomics on a scalable infrastructure. Nucleic Acids Res. 49 (W1), W185–W192.
Furfaro, L.L., Chang, B.J., Payne, M.S., 2018. Perinatal Streptococcus agalactiae epidemiology and surveillance targets. Clin. Microbiol. Rev. 31 (4) (p. e00049–18).
Gori, A., et al., 2020. Pan-GWAS of *Streptococcus agalactiae* highlights lineage-specific genes associated with virulence and niche adaptation. mBio (p. e00728–20).
Grant, J.R., Stothard, P., 2008. The CGView server: a comparative genomics tool for circular genomes. Nucleic Acids Res. 36 (suppl_2), W181–W184.
Gu Hao, L.Y., Jin, Zhang, Ying, Wang, Zhiyong, Liu, Ping, Cheng, Xingyong, Wang, Quanming, Zou, Jiang, Gu, 2018. Rational design and evaluation of an artificial Escherichia coli K1 protein vaccine candidate based on the structure of OmpA. Frontiers in cellular and infection. Microbiology 8.
Hua, X., et al., 2021. BacAnt: a combination annotation server for bacterial DNA sequences to identify antibiotic resistance genes, integrons, and transposable elements. Front. Microbiol. 12.
Kamruzzaman, M., Wu, A.Y., Iredell, J.R., 2021. Biological functions of type II toxin-antitoxin systems in bacteria. Microorganisms 9 (6), 1276.
Khalessi, N., Afsharkhas, L., 2014. Neonatal meningitis: risk factors, causes, and neurologic complications. Iran. J. Child Neurol. 8 (4), 46–50.
Kutschera, V.E., van der Valk, T., et al., 2022. GenErode: a bioinformatics pipeline to investigate genome erosion in endangered and extinct species. BMC Bioinforma. 23, 228.
Langille, M.G., Hsiao, W.W., Brinkman, F.S., 2008. Evaluation of genomic island predictors using a comparative genomics approach. BMC Bioinforma. 9 (1), 1–10.
Lannes-Costa, P.S., et al., 2020. Comparative genomic analysis and identification of pathogenicity islands of hypervirulent ST-17 Streptococcus agalactiae Brazilian strain. Infect. Genet. Evol. 80, 104195.
Liu, B., et al., 2019. VFDB 2019: a comparative pathogenomic platform with an interactive web interface. Nucleic Acids Res. 47 (D1), D687–D692.
Lynskey, N.N., et al., 2019. RocA binds CsrS to modulate CsrRS-mediated gene regulation in group a Streptococcus. Mbio 10 (4) (p. e01495–19).
Matsue, M., et al., 2020. Pathogenicity characterization of prevalent-type Streptococcus dysgalactiae subsp. equisimilis strains. Front. Microbiol. 11, 97.
Michaud, J., et al., 2008. Integrative analysis of RUNX1 downstream pathways and target genes. BMC Genomics 9 (1), 1–17.
Moulin, P., et al., 2016. The Adc/Lmb system mediates zinc acquisition in Streptococcus agalactiae and contributes to bacterial growth and survival. J. Bacteriol. 198 (24), 3265–3277.
Neely, M.N., Wiafe-Kwakye, C.S., Molloy, S.D., 2020. Determining the role of bacteriophages in the virulence of Streptococcus agalactiae. Multidiscip. Digital Publ. Inst. Proc. 50 (1), 20.
Noble, K., et al., 2021. Group B Streptococcus cpsE is required for serotype V capsule production and aids in biofilm formation and ascending infection of the reproductive tract during pregnancy. ACS Infect. Dis. 7 (9), 2686–2696.
Palang, I., et al., 2020. Cytotoxicity of Streptococcus agalactiae secretory protein on tilapia cultured cells. J. Fish Dis. 43 (10), 1229–1236.
Palmer, M., et al., 2020. All ANIs are not created equal: implications for prokaryotic species boundaries and integration of ANIs into polyphasic taxonomy. Int. J. Syst. Evol. Microbiol. 70 (4), 2937–2948.
Radtke, A., et al., 2010. Rapid multiple-locus variant-repeat assay (MLVA) for genotyping of Streptococcus agalactiae. J. Clin. Microbiol. 48 (7), 2502–2508.
Rantala, S., 2014. Streptococcus dysgalactiae subsp. equisimilis bacteremia: an emerging infection. Eur. J. Clin. Microbiol. Infect. Dis. 33 (8), 1303–1310.
Silvestre, I., Borrego, M.J., Jordão, L., 2020. Biofilm formation by ST17 and ST19 strains of Streptococcus agalactiae. Res. Microbiol. 171 (8), 311–318.
Sridharan, U., et al., 2019. Molecular dynamics simulation of metal free structure of Lmb, a laminin-binding adhesin of Streptococcus agalactiae: metal removal and its structural implications. J. Biomol. Struct. Dyn. 37 (3), 714–725.
Stewart, A.G., et al., 2020. Whole genome sequencing for antimicrobial resistance mechanisms, virulence factors and clonality in invasive Streptococcus agalactiae blood culture isolates recovered in Australia. Pathology 52 (6), 694–699.
Tettelin, H., et al., 2002. Complete genome sequence and comparative genomic analysis of an emerging human pathogen, serotype V Streptococcus agalactiae. Proc. Natl. Acad. Sci. 99 (19), 12391–12396.
Thomas, L., Cook, L., Richardson, A.R., 2020. Two-component signal transduction systems in the human pathogen Streptococcus agalactiae. Infect. Immun. 88 (7) (e00931–19).
Torkian, B., et al., 2020. BLAST-QC: automated analysis of BLAST results. Environ. Microbiome 15 (1), 1–8.
Uchiyama, S., et al., 2019. Dual actions of group B Streptococcus capsular sialic acid provide resistance to platelet-mediated antimicrobial killing. Proc. Natl. Acad. Sci. 116 (15), 7465–7470.
Urgent Need for Vaccine to Prevent Deadly Group B streptococcus, 2021 (London).
Vasi, J., Lindberg, M., Guss, B., 2000. A novel IS-like element frequently inserted in a putative virulence regulator in bovine mastitis isolates of Streptococcus dysgalactiae. Plasmid 44 (3), 220–230.
Wang, R., et al., 2018. Corrigendum: pathogenicity of human ST23 Streptococcus agalactiae to fish and genomic comparison of pathogenic and non-pathogenic isolates. Front. Microbiol. 9, 173.
WHO, 2021. Meningitis. WHO, Geneva.
do Nascimento, C.S., et al., 2019. Streptococcus agalactiae in pregnant women in Brazil: prevalence, serotypes, and antibiotic resistance. Braz. J. Microbiol. 50 (4), 943–952.